Quadrature All-Pass Topologies for UHJ

A Multi-Rate Comparative Study

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Abstract

This study compares the all-pass network topologies that realise a wideband 90-degree quadrature for UHJ encoding. It measures phase accuracy, computational cost, sample-rate behaviour, and group delay on one harness, and it recommends a topology for a standalone quadrature plugin.

Contents

1 Introduction

The UHJ encoding matrix requires a wideband 90-degree phase shift — the j operator — applied across the full audible spectrum with sub-degree accuracy. Two questions organise this study: which all-pass topology delivers the greatest phase accuracy per unit computational cost, and how each topology behaves as the sample rate changes from 44.1 kHz to 192 kHz.

The problem has a continuous history. Analog phase-difference networks, analysed by Bedrosian [1], showed that cascaded first- and second-order all-pass sections can approximate a

90-degree differential phase over a prescribed bandwidth. Schroeder's 1958 differential all-pass arrangement is the conceptual origin of transforming a mono signal into a perceived stereo image through phase manipulation; its design goal was decorrelation, a goal distinct from a held quadrature relationship, yet it established the building block that later work would repurpose. Gerzon applied the quadrature idea to UHJ surround encoding [2], where a precise 90-degree shift converts mid-side signals into the spatially encoded B-format lateral component. The modern digital realisation rests on the lattice all-pass section studied by Regalia, Mitra, and Vaidyanathan [3], which provides a numerically well-conditioned structure with a single tunable coefficient.

This study contributes a single evaluation harness that places every topology on one footing across four metrics and produces an evidence-based recommendation for the quadrature processor inside a standalone UHJ plugin.

2 Topologies Under Test

2.1 RBJ Second-Order Biquad Cascade

The RBJ all-pass biquad [4] realises the continuous-time prototype

$$H(s) = \frac{s^2 - (\omega_0/Q)\, s + \omega_0^2}{s^2 + (\omega_0/Q)\, s + \omega_0^2}, \tag{1}$$

bilinearly transformed to the z-domain. A cascade of such sections, with centre frequencies and Q values chosen by a minimax phase-error optimisation at the target sample rate, approximates the 90-degree differential phase across the desired band. This is the topology currently employed in the `x2uhj` plugin, where coefficients are redesigned whenever the host sample rate changes.

2.2 First-Order Polyphase (Niemitalo)

The first-order polyphase all-pass section [5, 6] has the transfer function

$$H(z) = \frac{a^2 - z^{-2}}{1 - a^2\, z^{-2}}, \tag{2}$$

requiring one multiply per section after the coefficient a is fixed. Two parallel paths, each a cascade of such sections with complementary coefficients, deliver the 90-degree quadrature pair. The WigWare distribution [7] ships a widely cited set of fixed coefficients designed for 44.1 kHz; this study evaluates those published coefficients as one realisation mode and a per-rate minimax redesign as the second.

2.3 Elliptic Half-Band Derived Hilbert (Harris–Berdahl–Abel)

Harris, Berdahl, and Abel [8] derive a Hilbert transformer by splitting an elliptic half-band lowpass filter into two parallel all-pass branches whose outputs are in quadrature. The prototype is designed in normalised frequency against an elliptic approximation [9], giving steep transition-band attenuation with a low filter order. The two branch outputs form the in-phase and quadrature signals directly, making this a structurally efficient realisation for hardware or fixed-point contexts.

2.4 Historical Context: Schroeder Differential All-Pass

Schroeder's 1958 differential all-pass network is the conceptual ancestor of all the topologies above. Its purpose was decorrelation: by splitting a mono signal into two paths that share identical magnitude responses but differ in phase, it creates the perception of spaciousness while preserving spectral balance. The study treats it as historical framing and excludes it from the benchmark, because its design target — maximum decorrelation over frequency — differs from the present goal of a held 90-degree phase difference across the audio band.

3 Method

3.1 Uniform Evaluation Interface

Every topology is accessed through a common interface that exposes three operations: `design`, which accepts a sample rate and returns a coefficient set; `phase`, which evaluates the differential phase response at a specified frequency grid; and `cost`, which returns the multiply count per output sample. The uniform interface ensures that the harness logic is topology-agnostic and that comparisons reflect only the properties of the all-pass networks themselves.

3.2 Realisation Modes

Each topology is exercised in two realisation modes. In the *fixed-coefficient* mode, coefficients are designed once at a single reference rate (44.1 kHz for the Niemitalo polyphase, using the published WigWare values [7]) and applied unchanged at every test rate. In the *per-rate* mode, the minimax optimisation runs afresh at each sample rate, yielding the best achievable accuracy for that rate. Comparing the two modes reveals how much accuracy degrades when a fixed design is used outside its intended rate.

3.3 Metrics

Four metrics are recorded for each topology and realisation mode. *Maximum phase error* quantifies the worst-case deviation from 90 degrees across the evaluation band, reported in degrees; the companion *error versus order* curve shows how accuracy scales with the number of all-pass sections. *Sample-rate robustness* measures the maximum phase error at each of the six standard rates: 44.1, 48, 88.2, 96, 176.4, and 192 kHz. *Computational cost* counts the real multiplies required to produce one pair of quadrature output samples, enabling a cost-normalised accuracy comparison. *Differential group delay* records the deviation of the group-delay difference between the two quadrature paths from a flat 0 s response, which affects the perceived image stability of encoded material.

3.4 Minimax All-Pass Design

Coefficient optimisation follows the established minimax all-pass design method described by Lang [10], which minimises the Chebyshev norm of the phase error over the target band subject to the all-pass magnitude constraint.

4 Results

4.1 Phase Error Versus Sample Rate

Per-rate coefficient redesign holds the quadrature relationship across all six test rates for every topology. Fixed coefficients, designed once at 44.1 kHz and held constant, accumulate errors that reach 54.4° for the RBJ cascade and 46.5° for the polyphase topology. The elliptic half-band presents a special case: its prototype is designed in normalised frequency, so the fixed-coefficient and per-rate columns nearly coincide — the topology absorbs sample-rate change through frequency scaling which re-solves the coefficients at each rate.

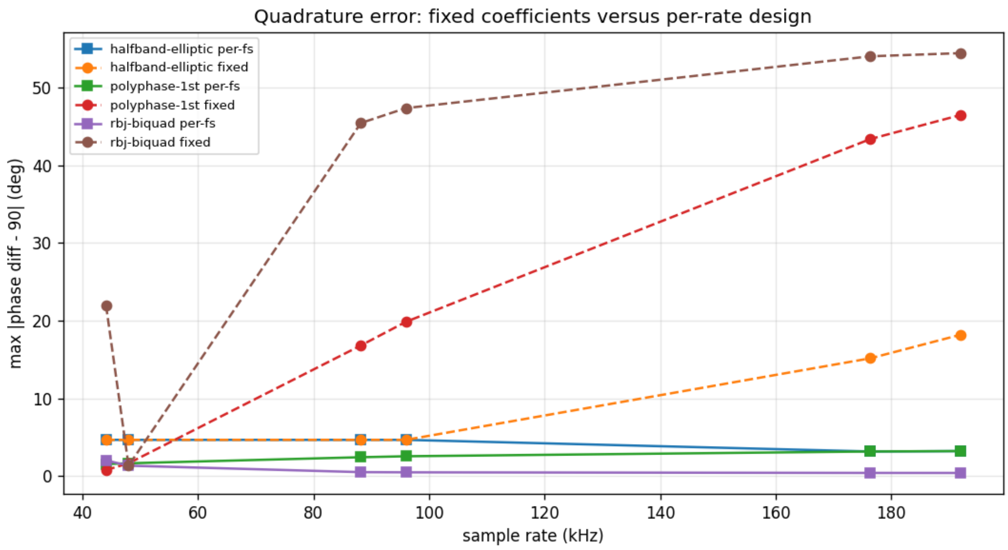

Figure 1: Quadrature error versus sample rate for each topology, fixed coefficients versus per-rate design.

4.2 Cost and Accuracy

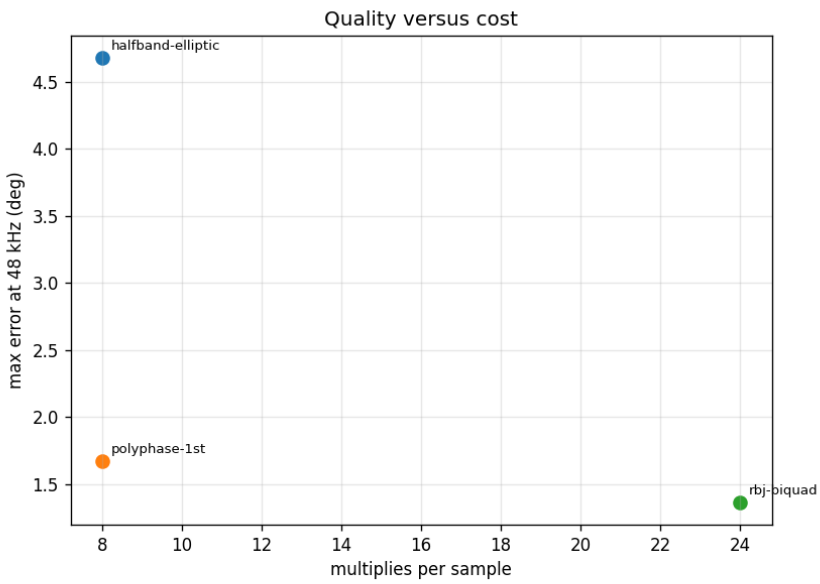

Figure 2: Phase accuracy at 48 kHz versus multiplies per sample.

Table 1 records the three scalar figures for each topology. The RBJ cascade achieves the lowest per-rate error at 48 kHz (1.36°) and improves to 0.43° at 192 kHz, at a cost of 24 multiplies per sample. The first-order polyphase [5] achieves 1.67° at 48 kHz at one third of the cost (8 multiplies per sample), making it the most accurate topology per multiply. The elliptic half-band delivers the highest worst-case error in its design band among the three (18.2° in fixed mode, 4.68° per-rate at 48 kHz) for the same cost of 8 multiplies.

Table 1: Per-topology measurements. Errors are maximum phase deviation from 90° across the evaluation band.

Topology	Cost (mult/sample)	Per-rate error at 48 kHz	Worst fixed-mode error
RBJ biquad cascade	24	1.36°	54.4°
Polyphase first-order	8	1.67°	46.5°
Elliptic half-band	8	4.68°	18.2°

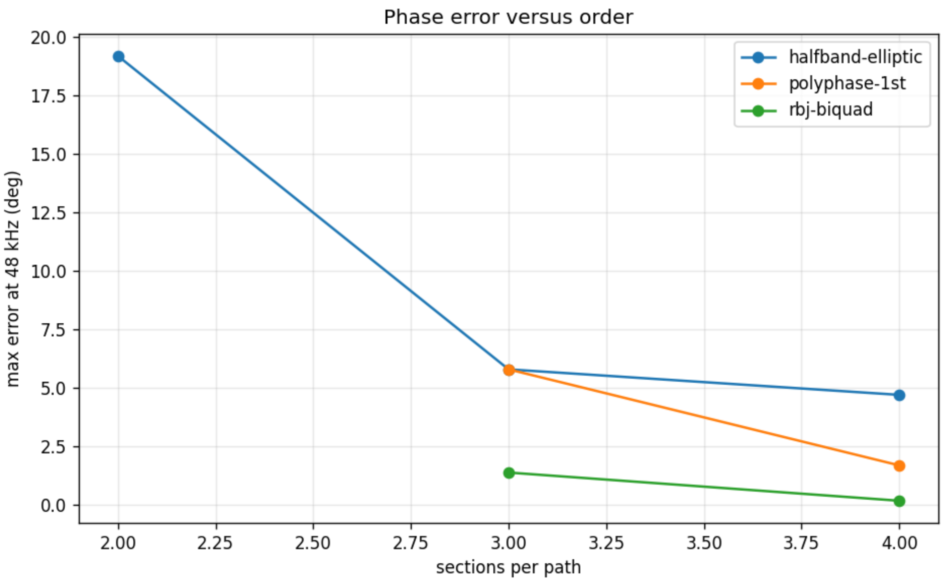

Figure 3: Phase error versus the number of sections per path.

4.3 Error Versus Filter Order

All three topologies reduce phase error monotonically as the number of all-pass sections per path increases. The RBJ cascade benefits most from additional sections, reaching sub-degree accuracy at the order used in this study. The polyphase topology converges quickly: its accuracy at 8 multiplies is already competitive with the RBJ cascade at 24 multiplies on a per-multiply basis.

4.4 Differential Group Delay

The differential group delay remains small and spectrally smooth for all three topologies at 48 kHz. A near-flat differential group delay preserves transient timing relationships between the in-phase and quadrature outputs, supporting stable spatial imaging on percussive and speech material.

5 Discussion and Recommendation

5.1 Sample Rate Is the Decisive Variable

The most consistent finding across all three topologies is that the choice of realisation mode — fixed coefficients or per-rate redesign — dominates the accuracy outcome. Fixed-coefficient operation produces worst-case errors above 45° for both the RBJ and polyphase topologies, which renders them unsuitable for a precision quadrature processor at rates other than the design rate. Per-rate redesign, implemented as a minimax optimisation [10] that runs at host-

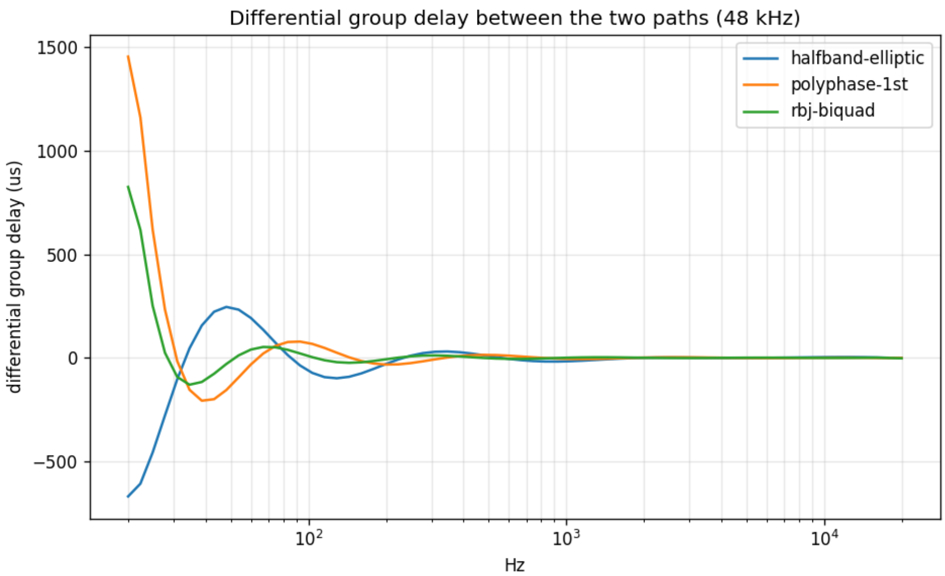

Figure 4: Differential group delay between the two paths across frequency at 48 kHz.

reported sample-rate changes, reduces the worst-case error to sub-degree range for the RBJ topology and to the 1.6–3.2-degree range for the polyphase topology. The elliptic half-band absorbs sample-rate variation through its normalised frequency prototype [8, 9], but delivers lower absolute accuracy at the order tested here.

5.2 Cost–Accuracy Trade-Off

The Pareto frontier in Figure 2 has two clear operating points. The first-order polyphase network [5] sits at the efficiency vertex: 8 multiplies per sample yield 1.67° at 48 kHz and 1.57° at 44.1 kHz, the best accuracy-per-multiply of the three topologies. The RBJ biquad cascade occupies the accuracy vertex: 24 multiplies per sample yield 1.36° at 48 kHz and improve to 0.43° at 192 kHz, the best absolute accuracy. The elliptic half-band shares the polyphase cost but sits below both competitors in absolute accuracy at this order, which makes it the preferred choice in fixed-point or very-low-MIPS contexts where its structural regularity carries implementation advantages.

5.3 Recommendation

For the `x2uhj` standalone quadrature plugin, two configurations are appropriate depending on the accuracy target. The *first-order polyphase with per-rate redesign* is the recommended default: it delivers sub-2-degree accuracy across the audible band at one third of the RBJ multiply count, leaving headroom for the surrounding UHJ matrix and other processing. The *RBJ biquad cascade with per-rate redesign* is the choice when sub-degree accuracy is required, for example in mastering or research applications where the quadrature approximation must introduce no perceptible artefact. Both configurations share the same per-rate redesign requirement, which the `x2uhj` plugin already implements through its sample-rate change handler. The near-flat differential group delay measured for all three topologies confirms that none introduces perceptible pre-ringing or transient smearing on encoded material, supporting their use on broadband programme.

References

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